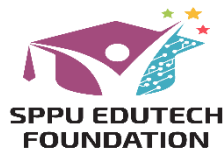


SPPU EDUTECH FOUNDATION

Savitribai Phule Pune University, Pune-411007



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SPPU EDUTECH FOUNDATION offers education services, training services, mentoring services, consulting service in the areas of Software Development, IT Infrastructure, Information Security, System Integration, IT Infrastructure Management, Application Management, Data Centre Management, Training & Educational Content & Platform Development etc.

Savitribai Phule Pune University is one of the leading universities of India, which also leads in overall automation across its all Administrative process. Since the first Process automation Project of Affiliation Online in 2008, University has come long way with application development for Administration, Finance, Examination, Board of Deans, University Campus etc.

Technical Integration Document

Technical Integration Document for Online Payment Gateway Integration

1. Challan Generation Logic

The Challan Generation process must strictly follow the steps below:

1.1 Challan Number Format

- Year (2 digits): e.g., 25 (for 2025)
- Process Code (2 digits): e.g., 90
- Month (2 digits): Current month number (e.g., 06 for June)
- Auto Increment (5 digits): Starts from 00001, increments with each challan.

Final Challan Format (11 Digits):

<Year(2)> + <Process Code(2)> + <Month(2)> + <Auto Increment(5)>

Year:25

Process code: 87

Month: 37 : 36+1 (1 means January, same process as next month)

Auto increment: last 5 digits

Total: 11-digit Challan number required.

.Example: 25873700001

1.2 SQL Function to Generate Challan Number

```
CREATE FUNCTION [FC].[FC_Challan_No_Generator] (@Process_Code VARCHAR(2))
RETURNS VARCHAR(15)
AS
BEGIN
    DECLARE @AutoID VARCHAR(100);
    DECLARE @Year CHAR(2) = FORMAT(GETDATE(), 'yy');
    DECLARE @Month CHAR(2) = FORMAT(GETDATE(), 'MM');
    DECLARE @ProcessCode CHAR(2) = @Process_Code;
    DECLARE @AutoIncrement BIGINT;

    SET @Month = CAST(@Month AS INT) + 12;

    SELECT @AutoIncrement = ISNULL((
        SELECT LEFT(RIGHT(MAX(Challan_no), 5), 5)
        FROM TableName
        WHERE Challan_no LIKE @Year + @ProcessCode + @Month + '%' AND LEN(Challan_no) = 11
```

```
), 0);  
  
SET @AutoIncrement = @AutoIncrement + 1;  
  
SET @AutoID = RIGHT('00000' + CAST(@AutoIncrement AS VARCHAR), 5);  
  
RETURN @Year + @ProcessCode + @Month + @AutoID;  
END
```

Service Overview

- Service Name: Finance Challan Submission Service
 - Service URL: <https://mkclreconcile.unipune.ac.in/Service.svc>
 - Purpose: Send the challan information to the Finance Department for reconciliation.
 - Protocol: SOAP/HTTP
 - Authentication: Encrypted Static Token required for each request.
-

1. Integration in ASP.NET (C#)

1.1 Service Reference

- Add Service Reference to your project with URL: <https://mkclreconcile.unipune.ac.in/Service.svc>
 - Namespace Example: ServiceReference1
-

1.2 ASP.NET Code Example

```
public void Finance_ChallanService()
{
    #region "Finance Challan Web Service Integration"

    ServiceReference1.ServiceClient Sendch1 = new ServiceReference1.ServiceClient();

    DataSet ds1 = new DataSet();
    SqlCommand cmd = new SqlCommand
    {
        CommandText = "HAS.International_Challan_Service",
        CommandType = CommandType.StoredProcedure
    };
    cmd.Parameters.AddWithValue("@Mode", "PermanentChallan");
    cmd.Parameters.AddWithValue("@Application_ID", Convert.ToString(Session["Application_ID"]));

    dbcon ObjDbcon = new dbcon();
    ObjDbcon.ExecuteDataset(cmd, ref ds1);

    if (ds1.Tables[0].Rows[0][0].ToString() != "")
    {
        string challanXml = ds1.Tables[0].Rows[0][0].ToString();
        string challanno = ds1.Tables[1].Rows[0][0].ToString();

        string key = "d709ie08dn2dg4201fi1"; // Encryption Key
        string StaticToken = EncodePasswordToBase64(key);

        Sendch1.ImportChallanXMLNew(StaticToken, challanno, challanXml);
    }

    #endregion
}
```

```
}
```

1.3 Encryption Method for Static Token

```
public static string EncodePasswordToBase64(string key)
{
    try
    {
        byte[] encData_byte = Encoding.UTF8.GetBytes(key);
        return Convert.ToBase64String(encData_byte);
    }

    catch (Exception ex)
    {
        throw new Exception("Error in base64Encode: " + ex.Message);
    }
}
```

2. Database Side (SQL Server)

2.1 Stored Procedure

Procedure Name: HAS.International_Challan_Service

Parameter: @Mode = 'PermanentChallan'

Example Logic:

```
IF (@Mode = 'PermanentChallan')
BEGIN
    SELECT
        (
            SELECT
                PACD.Challan_No AS ChallanNo,
                PACD.Application_ID AS ApplicationNumber,
                " AS facultyCode,
                PACD.Bank_Type AS bankName,
                CASE WHEN PACD.Bank_Type = 'BANK OF MAHARASHTRA' THEN 1 ELSE 2 END AS bankCode,
                '2016-2017' AS Acad_Year,
                CASE WHEN PACD.Challan_Type = 'Offline' THEN 'CASH' ELSE 'ONLINE' END AS amtType,
                ISNULL(PACD_In.Refund_Deposit, 0) AS AMOUNT,
                (
                    SELECT '400002' AS AccountID,
                        ISNULL(PACD_In.NonRefund_Deposit, 0) AS AMOUNT,
                        'C' AS DBCRType,
                        1 AS StudCount
                    FROM HAS.International_Hstl_Challan_Details
                    WHERE Application_ID = PAD.Application_ID AND IsCurrent = 0
                    FOR XML PATH('GL'), TYPE
                ) AS GLAmt,
                (
                    SELECT
                        Nm.First_Name AS FirstName,
```

```

        Nm.Middle_Name AS MiddleName,
        Nm.Last_Name AS SurName,
        " AS MotherName
    FROM HAS.International_Hstl_Admission_Details Nm
    WHERE Nm.Application_ID = PAD.Application_ID
    FOR XML PATH(""), TYPE
) AS Name,
...
FOR XML PATH('Challan'), TYPE
) AS ChallanXML;

SELECT Challan_No
FROM HAS.International_Hstl_Challan_Details
WHERE Application_ID = @Application_ID AND IsCurrent = 0;
END;

```

3. Sample Challan XML Format

```

<Challan>
  <ChallanNo>25401012345</ChallanNo>
  <ApplicationNumber>11001045678</ApplicationNumber>
  <facultyCode></facultyCode>
  <bankName>HDFC</bankName>
  <bankCode>2</bankCode>
  <Acad_Year>2024-25</Acad_Year>
  <amtType>Cash</amtType>
  <AMOUNT>1700</AMOUNT>

  <GLAmt>
    <GL>
      <AccountID>105010</AccountID>
      <AMOUNT>500.00</AMOUNT>
      <DBCRTType>C</DBCRTType>
      <StudCount>1</StudCount>
    </GL>
    <GL>
      <AccountID>111028</AccountID>
      <AMOUNT>1200.00</AMOUNT>
      <DBCRTType>C</DBCRTType>
      <StudCount>1</StudCount>
    </GL>
  </GLAmt>

  <Student>
    <Name>
      <FirstName>TWAIN</FirstName>
      <MiddleName>VALERIAN</MiddleName>
      <SurName>GONSALVES</SurName>
      <MotherName>TULIP</MotherName>
    </Name>
    <emailId>example@gmail.com</emailId>
    <PRNNo>1061000238</PRNNo>
    <AddressDet>
      <Address>KUMAR PINNACLE A-1/406 TADIWALA ROAD</Address>
    </AddressDet>
  </Student>
</Challan>

```

```

    <City>PUNE</City>
    <Dist>PUNE</Dist>
    <State>Maharashtra</State>
    <Pin>411001</Pin>
  </AddressDet>
</Student>

<Collage>
  <Info>
    <CollegeName>ABC College of Engineering</CollegeName>
    <CollegeCode>1042</CollegeCode>
  </Info>
</Collage>

<Status>3</Status>
<userid>1</userid>
<EntryDate>10-Jun-2025</EntryDate>
</Challan>

```

4. Important Points

Field	Format	Notes
ChallanNo	11 Digits	2 (Year) + 2 (Code) + 7 (Sequence)
Application ID	11 Digits	Starts with 2-digit Department Code
bankCode	Numeric (1 or 2)	1 = Bank of Maharashtra; 2 = Others
amtType	Cash / Online	Based on payment type
Amount	Decimal	Total Amount Paid
EntryDate	dd-MMM-yyyy	UTC format preferred

5. Response Handling

- Web Service Method: ImportChallanXMLNew(string StaticToken, string challanNo, string challanXML)
-

2. Online Payment Service Integration

2.1 Service URL

<https://ops.unipune.ac.in/sppupgway/sppupgway.asmx>

2.2 Method: GetPaymentDetails()

XML Format for Request:

```
<Challan>
  <ApplicationName>Election Fund</ApplicationName>
  <ApplicationID>317000067</ApplicationID>
  <ChallanNo>17430000180</ChallanNo>
  <FullName>kale ashok mahadev</FullName>
  <MobileNo>7709191343</MobileNo>
  <EmailID>kaleaaaa@gmail.com</EmailID>
  <Amount>20</Amount>
  <Address1>pune</Address1>
  <Address2></Address2>
  <City></City>
  <Pincode>410503</Pincode>
  <State></State>
  <Country></Country>
  <ChallanDate>2017-06-21T14:12:25.117</ChallanDate>
  <IPAddress></IPAddress>
</Challan>
```

Note: Tag names must be exactly as given.

2.3 Redirection to Payment Gateway

After calling the method, redirect user to:

https://ops.unipune.ac.in/sppupgway/default.aspx?Application_ID=<Encrypted ChallanNo>

Example (in C#):

```
Response.Redirect(string.Format("https://ops.unipune.ac.in/sppupgway/default.aspx?Application_ID={0}",
    EncryptQueryString(HttpUtility.UrlDecode(challanNo.ToString()))));
```

2.4 Encryption/Decryption Logic (C#)

```
public string EncryptQueryString(string clearText)
{
    string EncryptionKey = "MAKV2SPBNI99212";
    byte[] clearBytes = Encoding.Unicode.GetBytes(clearText);
    using (Aes encryptor = Aes.Create())
```



```

{
    var pdb = new Rfc2898DeriveBytes(EncryptionKey, new byte[] { 0x49, 0x76, 0x61, 0x6e,
        0x20, 0x4d, 0x65, 0x64, 0x76, 0x65, 0x64, 0x65, 0x76 });
    encryptor.Key = pdb.GetBytes(32);
    encryptor.IV = pdb.GetBytes(16);
    using (var ms = new MemoryStream())
    {
        using (var cs = new CryptoStream(ms, encryptor.CreateEncryptor(), CryptoStreamMode.Write))
        {
            cs.Write(clearBytes, 0, clearBytes.Length);
            cs.Close();
        }
        clearText = Convert.ToBase64String(ms.ToArray());
    }
}
return clearText;
}

public string DecryptQueryString(string cipherText)
{
    string EncryptionKey = "MAKV2SPBNI99212";
    cipherText = cipherText.Replace(" ", "+");
    byte[] cipherBytes = Convert.FromBase64String(cipherText);
    using (Aes encryptor = Aes.Create())
    {
        var pdb = new Rfc2898DeriveBytes(EncryptionKey, new byte[] { 0x49, 0x76, 0x61, 0x6e,
            0x20, 0x4d, 0x65, 0x64, 0x76, 0x65, 0x64, 0x65, 0x76 });
        encryptor.Key = pdb.GetBytes(32);
        encryptor.IV = pdb.GetBytes(16);
        using (var ms = new MemoryStream())
        {
            using (var cs = new CryptoStream(ms, encryptor.CreateDecryptor(), CryptoStreamMode.Write))
            {
                cs.Write(cipherBytes, 0, cipherBytes.Length);
                cs.Close();
            }
            cipherText = Encoding.Unicode.GetString(ms.ToArray());
        }
    }
    return cipherText;
}

```

3. Web Service for Payment Response

3.1 Create Web Service Endpoint

- Accepts: XML string (POST)
- Example Response (XML Format):

```
<Challan>
<MihPayID>E2506270I948SS</MihPayID>
<Amt>500.0</Amt>
<Transaction_Amount>500.0</Transaction_Amount>
<TxnID>30655927062025233115</TxnID>
<ProductInfo>Test</ProductInfo>
<BankCode>Debit Card</BankCode>
<ApplicationID>RT108A030000515</ApplicationID>
<ChallanNo>25340005159</ChallanNo>
<FullName>Malekar Pushpa Vishwanath</FullName>
<MobileNo>9767515298</MobileNo>
<EmailID>malekarpushpa123@gmail.com</EmailID>
<Field9>success</Field9>
<Error_Code></Error_Code>
<Error_Message>Successful Transaction</Error_Message>
<Status>success</Status>
<UnMappedStatus>captured</UnMappedStatus>
<TranDate>2025-06-27 18:01:19</TranDate>
<BankType>Bank of Maharashtra</BankType>
</Challan>
```

3.2 Posting Payment Response to University Server

- Application Name (must match <ApplicationName> tag).
- Web Service URL:

<http://app1.unipune.ac.in/ele/ele/Webservices/PGReceivedChallan.asmx>

- Method Name:

GetPGReceivedDetails

3.3 Return URL (Post Transaction Completion)

Redirect to:

<http://election.unipune.ac.in/Registration/PaymentDetails.aspx>

This URL should handle both success and failure responses.

4. Summary of URLs and Methods

4.1 for example URL link

Purpose	URL
Service Reference	https://ops.unipune.ac.in/sppupgway/sppupgway.asmx
Payment Redirect	https://ops.unipune.ac.in/sppupgway/default.aspx
Response WebService	http://app1.unipune.ac.in/ele/ele/Webservices/PGReceivedChallan.asmx
Post-Payment Page	http://election.unipune.ac.in/Registration/PaymentDetails.aspx

Send the Following Details by mail

Application Name – Application should be the same as written in the < Application Name > tag.

Webservice URL -

E.g, Election Fund Web Service URL E.g., <http://app1.unipune.ac.in/ele/ele/Webservices/PGReceivedChallan.asmx>

Method Name - GetPGReceivedDetails

Return URL- redirects to the application after the transaction, either successful or failed